UrbanAtlas: Satellite Imagery for Urban Analysis

B.Vijaylakshmi, K.Mamatha, Ch.Roshika, Sk.Tehazeeb

Under the esteemed guidance of

Ms. N. Sandhya

Assistant Professor

Bachelor of Technology
Department of Information Technology
BVRIT HYDERABAD College of Engineering for Women

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Overview

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Introduction

- UrbanAtlas aims to create an annotated satellite imagery dataset for Indian urban and semi-urban regions, covering buildings, roads, water bodies, and vegetation.
- Supports semantic segmentation for accurate land-use classification and sustainable urban planning.
- Addresses the gap in India-specific annotated datasets despite high-resolution imagery availability.
- Objectives: Build an annotated dataset and compare it with satellite imagery from the past 10 years to analyze changes.

Stage-I Summary

- Collected a specific land-use dataset for the Hyderabad region, focusing on targeted area selection.
- Marked the dataset with geospatial markers to ensure accurate representation of land parcels.
- Used Labelbox for annotation and performed manual annotation of the Hyderabad dataset to establish ground truth.
- Conducted a detailed literature survey on various deep learning segmentation and detection models:
  - CNN architectures for feature extraction
  - U-Net for biomedical and spatial segmentation
  - DeepLab versions v1, v2, v3, v4 for semantic segmentation
  - PSPNet for pyramid scene parsing
- Compared strengths and limitations of these models in terms of accuracy, scalability, and computational efficiency.
- Based on the survey, YOLOv26 was chosen for its balance of speed, accuracy, and suitability for real-time urban land-use mapping tasks.

Implementation - Tech Stack (Stage-I)

AI & Logic	LLMs (OpenAI • Anthropic)
Browser Automation	Playwright
Dataset Collection	OpenStreetMap • Geospatial Markers
Annotation Tools	Labelbox • Manual Annotation (Hyderabad Dataset)
Model Survey	CNN • U-Net • DeepLab v1–v4 • PSPNet
Chosen Model	YOLOv26 (Speed & Accuracy for Real-Time Land Mapping)

Stage-I Outputs

Figure: Output 1

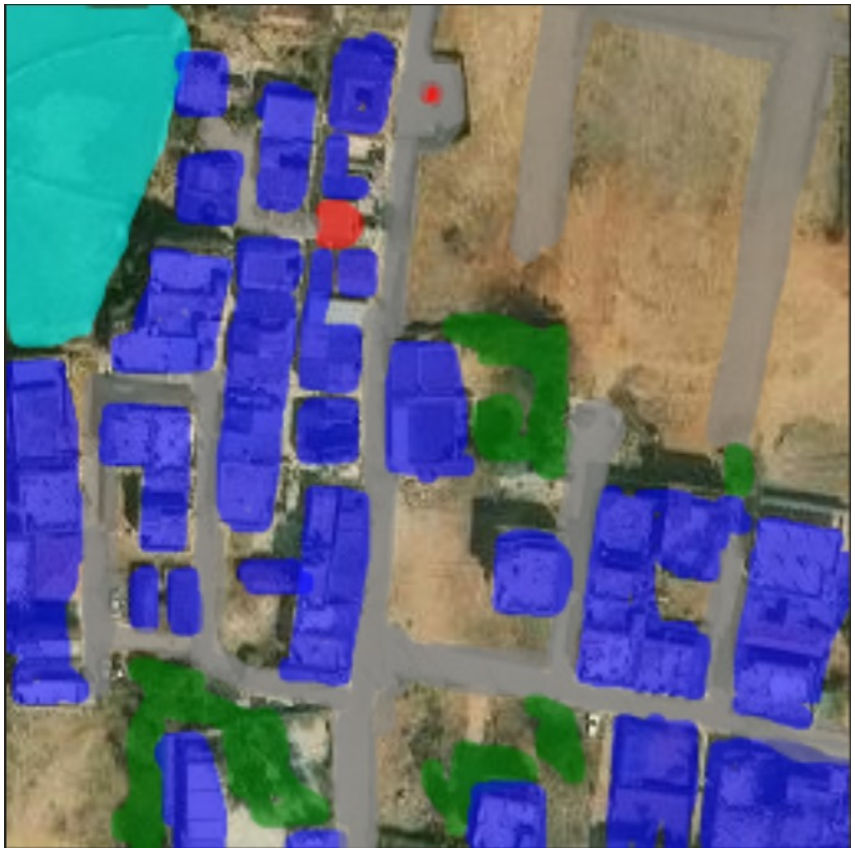
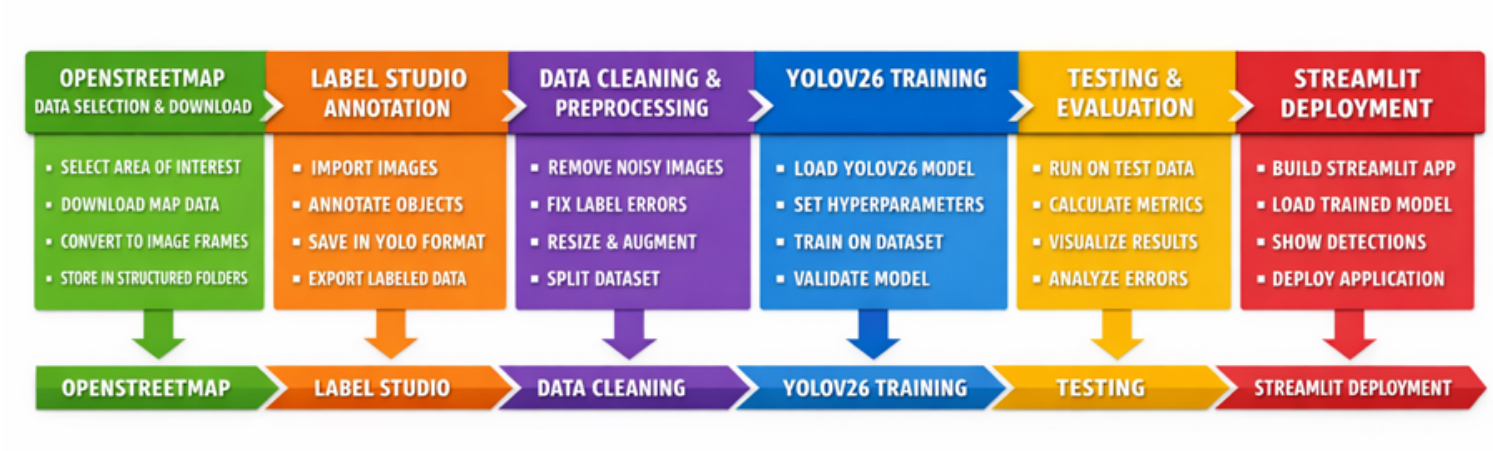


Figure: Output 2

Implementation - Architecture Diagram

- Data Collection Module (Satellite + OpenStreetMap)
- Annotation Module (Labelbox + Manual)
- Model Training Module (YOLOv26 with PyTorch/Ultralytics)
- Visualization Module (Plotly, Matplotlib, OpenCV)
- Deployment Module (Streamlit Interface)



Implementation - Tech Stack

Core Model	YOLOv26 (Ultralytics)
Frameworks	PyTorch • Ultralytics
Deployment & UI	Streamlit
Visualization Libraries	Matplotlib • Plotly
Computer Vision	OpenCV (opencv-python)
Package Management	pip (Python 3.13)

Modules / Interface

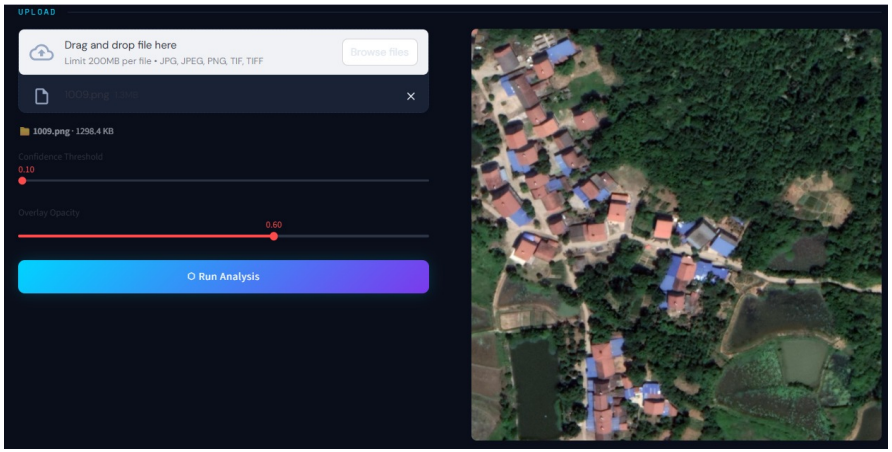
Module Integration

- Streamlit interface for user interaction
- Satellite + OpenStreetMap data ingestion
- Labelbox/manual annotation for Hyderabad dataset
- YOLOv26 (Ultralytics + PyTorch) for land-use detection
- Visualization with Plotly, Matplotlib, OpenCV

Interface

- Unified workflow: upload $\rightarrow$ annotate $\rightarrow$ train $\rightarrow$ visualize
- Real-time inference with clear land-use categories
- Interactive map outputs for urban analysis

Output Screens



Input

segmentation

Output Screens

Class details

Visualization

Conclusion

- UrbanAtlas successfully demonstrates an AI-driven land-use mapping system.
- Combines satellite imagery, OpenStreetMap data, and manual annotation for reliable ground truth.
- Employs YOLOv26 with PyTorch and Ultralytics for accurate urban segmentation.
- Streamlit interface enables real-time visualization and interactive exploration.
- Provides a scalable, reproducible workflow for urban analysis and smart city planning.

References

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- I. Dimitrovski, V. Spasev, S. Loshkovska, and I. Kitanovski, *U-net ensemble for enhanced semantic segmentation in remote sensing imagery*, Remote Sensing, vol. 16, no. 12, p. 2077, 2024. [Online]. Available: `https://doi.org/10.3390/rs16122077`
- Ultralytics, *YOLOv26 model documentation*, 2024.

Thank you